



Fatal *Rhodococcus erythropolis* endocarditis complicated by an abscess invading the atrioventricular node: a case report

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Abstract

Infective endocarditis (IE) and myocardial abscesses caused by *Rhodococcus erythropolis* are rare clinical manifestations in healthy individuals. We report the autopsy of a 43-year-old male who suffered from IE caused by *R. erythropolis* infection and died of an atrioventricular node abscess, arrhythmia, and heart failure. This case indicates that the possibility of infection affecting the cardiac conduction system should be considered in arrhythmia-related IE cases. Our findings suggest that clinicians should consider the possibility of *R. erythropolis* infection in arrhythmia cases associated with IE.

Keywords Infective endocarditis · Myocardial abscess · Atrioventricular node · *Rhodococcus erythropolis*

Introduction

Rhodococcus erythropolis is a Gram-positive bacterium widely distributed in nature, which is associated with various infections in immunocompromised hosts; however, only a few cases of human infection have been reported by previous studies [1, 2]. Here, we present an autopsy case of a patient who developed infective endocarditis (IE) caused by *R. erythropolis*, resulting in an abscess that invaded the

atrioventricular node (AVN), causing arrhythmia and heart failure. We conducted a detailed autopsy and pathological examination of the myocardial abscess following *R. erythropolis* infection.

Case report

Clinical history

A 43-year-old male was admitted to a regional clinic with a cough, expectoration, and shortness of breath. Blood tests revealed an elevated C-reactive protein level of 74.06 mg/L. Pneumonia was suspected, and the patient was treated with 10 mg dexamethasone and 1 g kanamycin, but there was no significant improvement in symptoms. The following day, at 9:00 am, his dyspnea worsened. He remained conscious, with a heart rate of 105 beats per minute, sinus tachycardia, a blood pressure of 120/60 mmHg, and a body temperature of 36.8 °C. Electrocardiography revealed frequent premature ventricular contractions (PVCs). The patient was then treated with amiodarone hydrochloride, levofloxacin hydrochloride tablets, and cefixime, and he requested discharge. However, at approximately 9:30 pm that evening, his condition worsened, ultimately resulting in death.

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Autopsy findings

Forensic necropsy was performed 48 h after death. The heart was enlarged (500 g) with mild thickening of the left ventricular wall. The aortic valve (AV) and mitral valve (MV) were conspicuously thickened and damaged by the formation of polypous and grayish vegetation. Upon examining the interventricular septum, a large off-white, soft mass (approximately $1.5 \times 1.0 \times 0.5$ cm) was found in the upper muscular portion of the septum adjacent to the posterior semilunar valve of the AV. The mass spanned almost the full thickness of the septal wall and was close to the left ventricular (LV) aspect, and the lesion extended to the central fibrous body (Fig. 1a and b). Microscopic examination revealed that the myocardial mass was infiltrated by extensive inflammatory cells, primarily neutrophils that were positive for myeloperoxidase (MPO). Additionally, there

was patchy myocardial liquefactive necrosis and fibroblast proliferation, consistent with a myocardial abscess (Fig. 1c and d). This purulent infection had extended to the area of the AVN with severe necrosis and fibrosis, as shown by hematoxylin and eosin (HE) and Masson-trichrome staining (Fig. 2a and b). HE staining showed necrosis and inflammatory cell infiltration in the mitral and tricuspid valve regions (Fig. 2c and d). Multiple patchy contraction band necrosis of the left ventricular myocardium was best seen with HE staining and Mallory's phosphotungstic acid hematoxylin (PTAH) staining (Fig. 2e and f).

Gram staining of the pericardial abscess area showed a Gram-positive coccal infection (Fig. 3a). For identification of the pathogenic species, rapid sequencing of the pathogenic microorganism was performed according to the next-generation sequencing (NGS) assay manual (BGI-Wuhan, China). Genomic DNA was extracted from

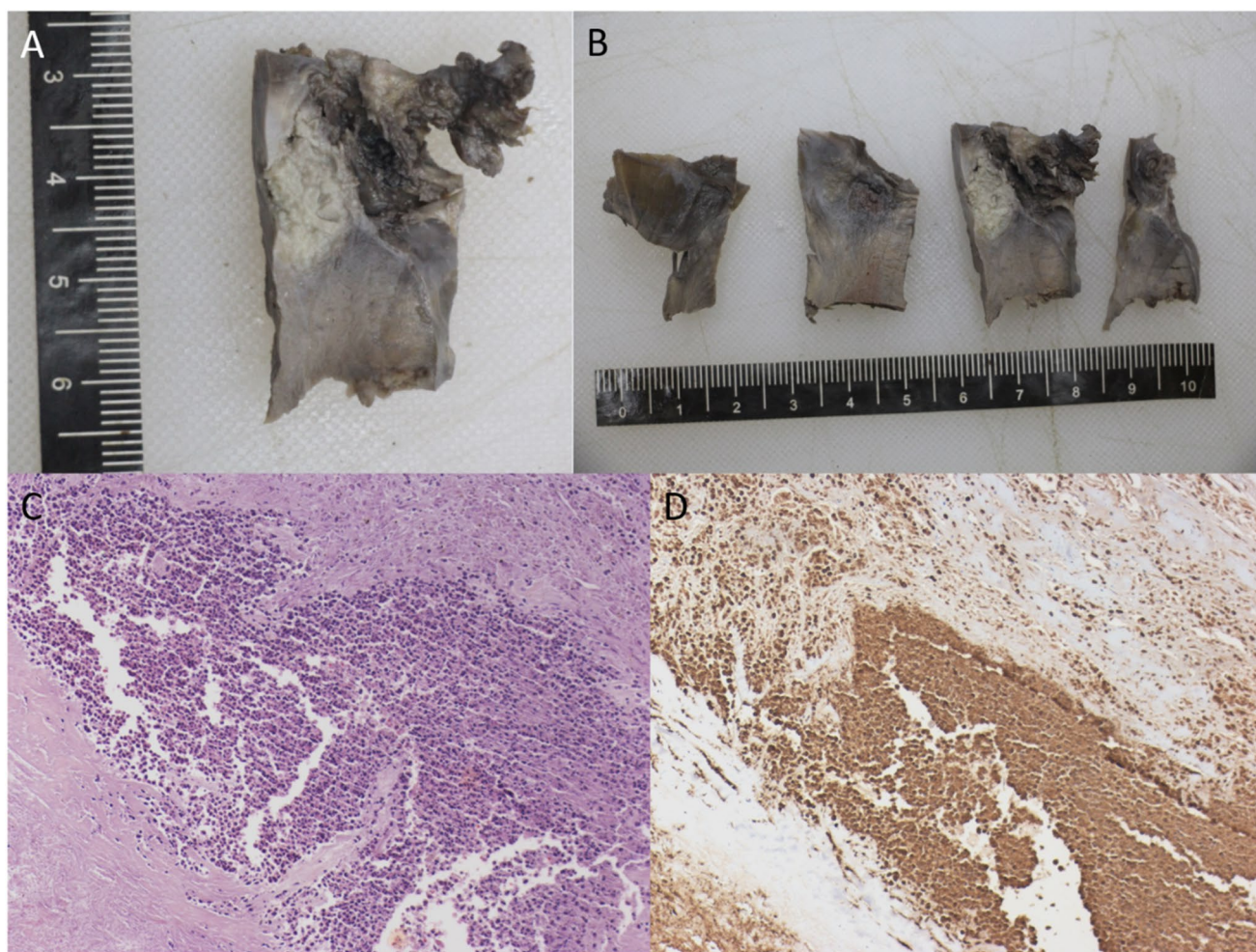


Fig. 1 Macroscopic findings of the interventricular septum. **(a)** The interventricular septum shows a large, soft, off-white mass. The mitral valve (MV) was markedly thickened and damaged by the formation of polypous and grayish vegetation. **(b)** Different sectional views of the abscess. **(c)** The abscess occupies the myocardial mass with extensive

inflammatory cell infiltration consisting mainly of neutrophils (hematoxylin and eosin (HE) stain, 100 $\times$). **(d)** Immunohistochemistry (IHC) of myeloperoxidase (MPO) was used to stain neutrophils of the section shown in panel C (IHC staining, 100 $\times$)

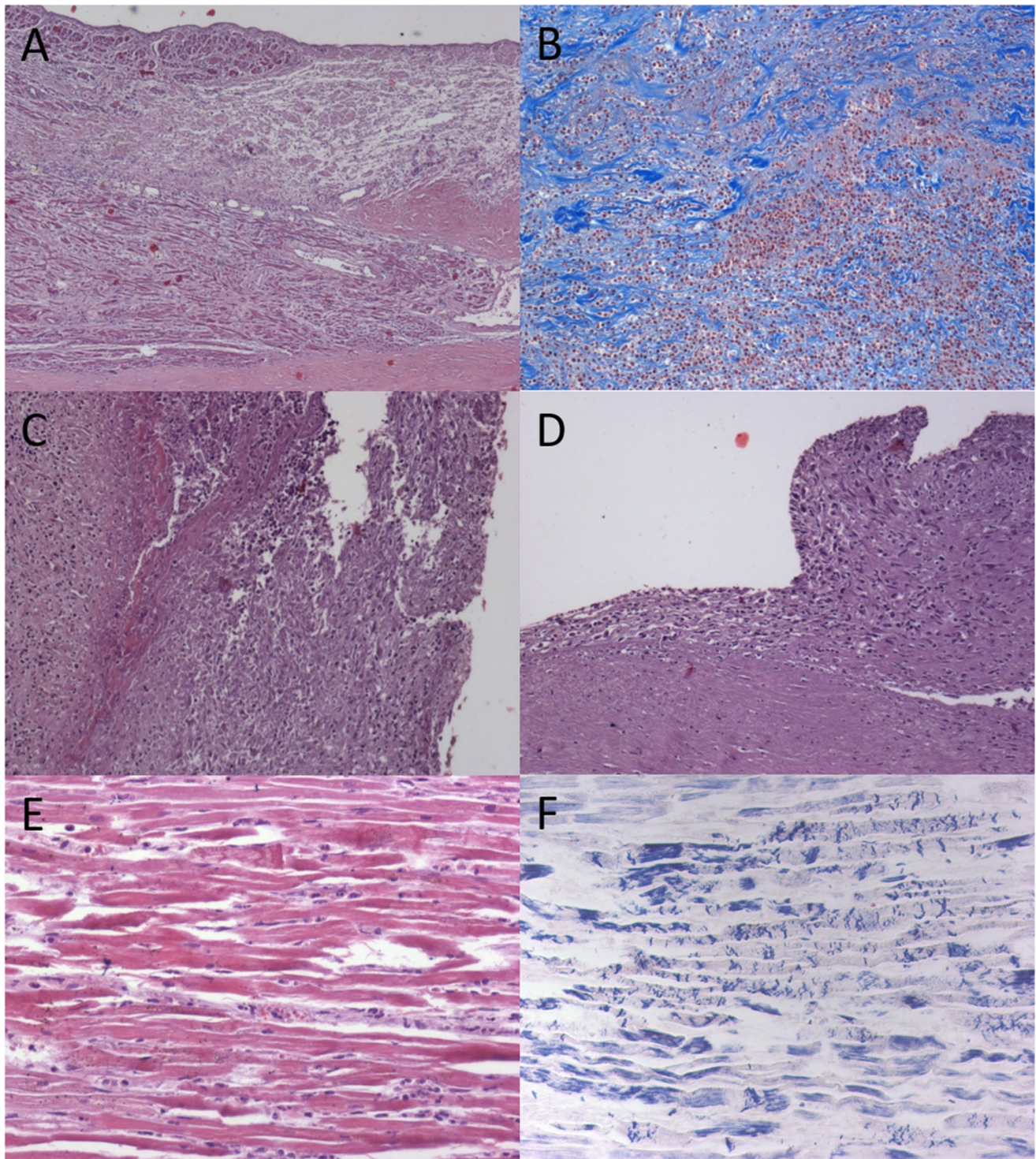


Fig. 2 (a) Hematoxylin and eosin (HE) staining shows necrosis and extensive infiltration of inflammatory cells in the atrioventricular node (AVN) area (HE stain, 40 $\times$). (b) Masson's trichrome staining was used to stain the AVN, which reveals severe necrosis with fibrosis and extensive infiltration of inflammatory cells (blue=fibrosis, red=inflam-

tory cells) (Masson's trichrome stain, 100 $\times$). (c and d) HE staining shows necrosis and inflammatory cell infiltration in the mitral valve (c) and tricuspid valve (d) regions (HE stain, 100 $\times$). (e and f) Necrosis in multiple patchy contraction bands was observed in the left ventricular myocardium (HE, 200 $\times$), (PTAH, 100 $\times$)

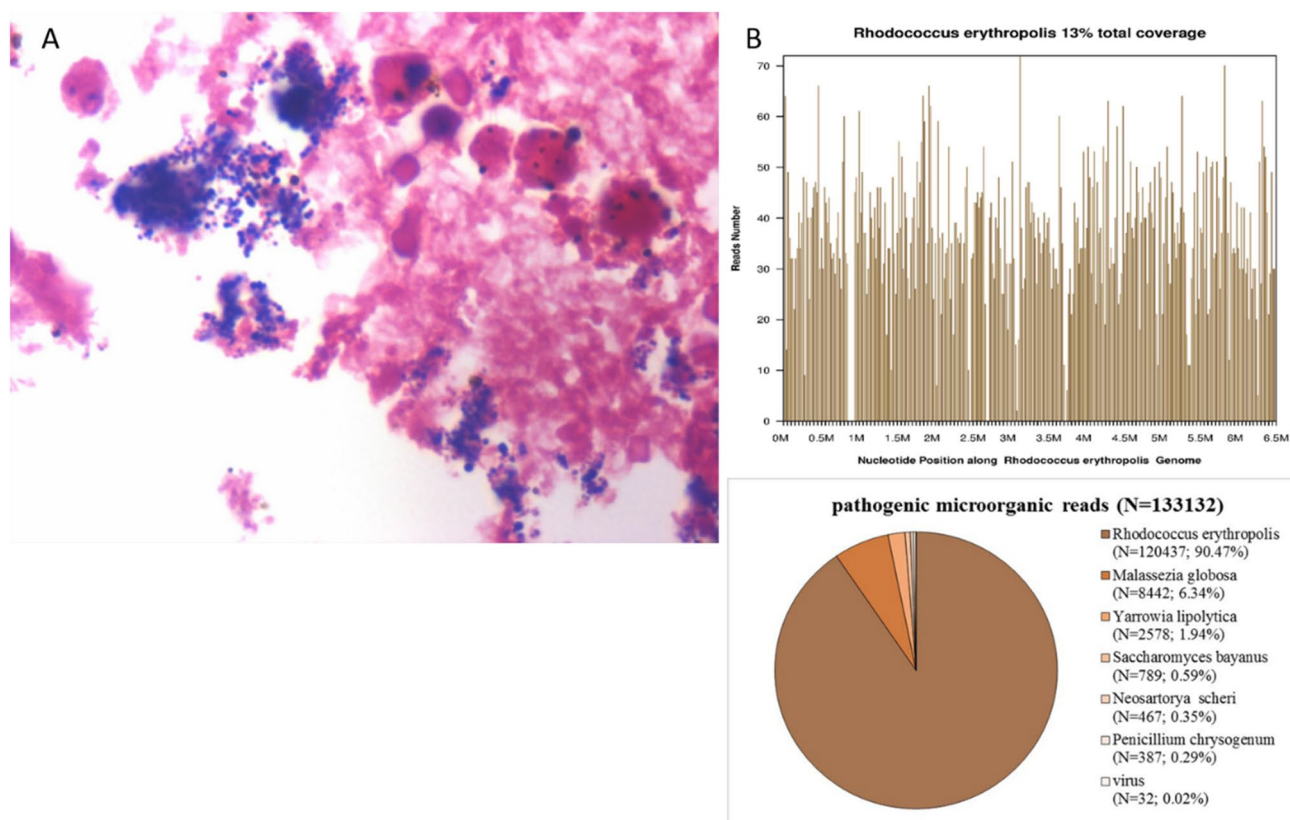


Fig. 3 (a) Gram staining of the pericardial abscess area showing a Gram-positive coccal infection (100×). (b) Next-generation sequencing results of pathogen identification in the cardiac tissue: 90.47% corresponded to *Rhodococcus erythropolis*, with a coverage of 13%

paraffin-embedded sections of the abscess areas using the MagPure FFPE DNA LQ kit (Magen, China). DNA libraries were constructed following end-repair, adapter ligation, and PCR, with sequencing on the BGISEQ-100 platform after quality control by the 2100 Bioanalyzer system (Agilent Technologies, Santa Clara, CA, USA). After removing human sequences, the remaining sequencing data were aligned with bacterial, viral, fungal, and protozoan databases. The mapped data were then processed for advanced analysis. As a result, a *R. erythropolis* subspecies was identified as the predominant pathogen in the inflammatory area of the heart, accounting for 90.47% (120,437 out of 133,132) of the total microorganism reads, with a coverage of 13% (Fig. 3b). This pathogen was not found in other organs.

Discussion

R. erythropolis infection in heart valves is rare, and most IE cases are caused by *Staphylococcus aureus* [3], making this case highly unique. *R. erythropolis* is a Gram-positive bacterium widely distributed in nature, particularly in soil. Although, it was previously considered of limited pathogenicity, there have been occasional clinical reports of human

infections. Fiori. et al. reported a case of *R. erythropolis*-induced heart prosthesis valve infection [4], and Shah et al. reported a case of aortic valve infection caused by *R. erythropolis* [5]. In addition, *R. erythropolis* has been linked to musculoskeletal infections, skin infections, sepsis, endophthalmitis, and meningitis [1]. In this case, we conducted a detailed autopsy and pathological analysis of the myocardial abscess caused by *R. erythropolis* infection.

Myocardial abscesses are rare but potentially fatal purulent infections that occur in approximately 20% of patients with IE [6]. In our case, the location of the abscess formation indicates that the myocardial abscess in the interventricular septum likely resulted from direct infection spread. The formation or diffusion of myocardial abscesses usually cause conduction abnormalities [7, 8], and frequent PVCs (defined as >1 PVC on a 10-second ECG or >30 PVCs per hour) are strongly associated with an increased risk of sudden cardiac death due to fatal cardiac arrhythmias [9]. In our case, severe arrhythmias (frequent PVCs) were detected prior to the patient's death, which may be linked to the spread of the myocardial abscess, affecting the AVN, and leading to its anatomical destruction. Furthermore, zonal contraction band necrosis in myocytes was observed in most areas of the heart, which could be related to episodes

of severe arrhythmias or systematic separate injuries to the myocardium.

Conclusion

In case of arrhythmia related to IE, the possibility of an infection affecting the cardiac conduction system should be considered. In addition, doctors should consider the possibility of *R. erythropolis* infection in cases of myocardial infection. When pathogen identification is challenging, NGS-based pathogen detection in paraffin-embedded sections of the abscess area is a valuable tool for diagnosing and identifying the causative pathogen in IE.

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Declarations

Compliance with ethical standards The case was approved by the ethics committee of Huazhong University of Science and Technology.

Informed consent Written informed consent was obtained from the family members of the patient.

Artificial intelligence statement We did not use any artificial intelligence tools to assist in writing.

Conflict of interest The authors declare that they have no conflict of interest.

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